NCAA March Madness 2026 Forecasting: A Two-Stage Machine Learning System for Tournament Selection and Bracket Outcomes (Men's + Women's)

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Abstract. Accurate forecasting of the NCAA March Madness tournament presents a dual-regime machine learning challenge. We propose a production-ready, two-stage system: **Task A: Selection Prediction** (team-season classification to determine the 68-team field) and **Task B: Bracket Outcome Prediction** (game-level probabilistic classification conditional on the field). The system is run separately for Men's (MBB) and Women's (WBB) basketball, with an empirical test of structural sharing. We employ a rigorous data governance strategy, strictly holding out the 2026 season for final out-of-sample evaluation and deploying layered defenses against data leakage, a known risk in sports forecasting. Evaluation across a broad family of models, including Decision Trees, MLPs, and kernel-based methods, led to the selection of a pooled Decision Tree model for Selection (AUC 0.9963) and separate models for Bracket outcomes: an MLP for MBB (AUC 0.8497) and an autoencoder-proxy MLP for WBB (AUC 0.8878). Our results confirm that Selection behaves as a high-signal threshold classification problem, while Bracket prediction is an inherently probabilistic problem, with WBB exhibiting a stronger hierarchy signal than MBB. The resulting pipeline produces a coherent set of 2026 forecast artifacts for both leagues.

1 Introduction

The NCAA Division I Basketball Tournaments (March Madness) are a major global event, with predictive modeling playing a significant role in media, analytics, and sports betting. Existing approaches often treat tournament prediction as a single, monolithic problem. However, the system is fundamentally two distinct tasks with different units of analysis and underlying statistical regimes: selecting the field of teams, and predicting game-by-game outcomes.

This paper presents an end-to-end Machine Learning pipeline designed to address this duality. Our primary contributions are:

- **Two-Stage Architecture**: Explicitly defining and modeling the selection (team-season classification) and bracket (game-level probabilistic classification) problems separately, demonstrating that this is the correct technical decomposition.

- **Rigorous Temporal Validation and Leakage Defense**: Implementing rolling multi-season cross-validation and a strict 2026 holdout to ensure model generalization and protect against common pitfalls like data leakage.

- **Empirical Comparison of Leagues**: Providing a side-by-side analysis of Men's (MBB) and Women's (WBB) leagues, confirming that the high-level discriminative structure for selection is shared (justifying pooling), but the probabilistic structure for bracket outcomes is different (justifying separate models).

2 System Architecture and Problem Decomposition

2.1 Task A: Selection Prediction (Team Season Classification)

This task aims to classify a team's season profile as either "Selected" or "Not Selected" into the tournament.

- **Unit of Analysis:** A single team season.
- **Label:** Win/Loss, with the primary output being a probability $P(Win)$.
- **Features:** Matchup differences, such as ELO_{diff}, $winpct_{diff}$, and differences in style/efficiency metrics. Absolute strength is encoded as a differential.
- **Regime:** The task is inherently **probabilistic**, requiring robust probability quality metrics and calibration due to the noise and possibility of upsets in single-game outcomes.

2.2 Task B: Bracket Outcome Prediction (Game-Level Probabilistic Classification)

This task predicts the win probability for a given matchup, conditional on the teams being in the field.

- **Unit of Analysis:** A single game matchup.
- **Label:** Win/Loss, with the primary output being a probability $P(Win)$.
- **Features:** Matchup differences, such as ELO_{diff}, $winpct_{diff}$, and differences in style/efficiency metrics. Absolute strength is encoded as a differential.
- **Regime:** The task is inherently **probabilistic**, requiring robust probability quality metrics and calibration due to the noise and possibility of upsets in single-game outcomes.

3 Experimental Design and Data Governance

3.1 Data Sources and Feature Provenance

We use historical NCAA data for MBB and WBB, explicitly separated to prevent schema mixing. Core data includes transactional game logs, team metadata, team-season aggregates, and time-derived ELO ratings. We maintain a feature provenance manifest to flag higher-risk features (e.g., *end_elo*) that could inadvertently capture post-decision information.

3.2 Validation Strategy and Leakage Defense (Overfitting Selection)

Evaluation correctness is paramount. We employ a **rolling multi-season validation** strategy to test generalization across different eras. The **2026 season** is designated as a strict, non-negotiable **holdout set**, excluded from all model training, hyperparameter tuning, and feature engineering decisions.

Key leakage defenses include the following:

- **As-of Dataset Versions:** Computing features at different time cutoffs to test for feature stability relative to the selection decision point.
- **Label Permutation Tests:** Shuffling the target labels and confirming that model performance (AUC) collapses to chance $\sim$, which supports no leakage from features to the label. We permute the labels and expect chance performance. We observe null AUC $\approx .5, which supports no leakage.$
- **Feature Manifest Audit:** We explicitly flag higher-risk features and treat them carefully.
- **Leakage Risk Scoring and Model Rejection:** We explicitly flag higher-risk features and treat them carefully.
- **Single-feature Smoke Tests:** We identify if a single feature dominates; dominance is not illegal but must be explained.

3.3 Evaluation Metrics

- **Selection (Task A):** AUC, PR-AUC, and **precision@K/recall@K**, as the business problem is a top-K (68-team) ranking decision.
- **Bracket (Task B):** AUC (for ranking), **log-loss, Brier Score, and Expected Calibration Error(ECE)**, which are necessary for simulation-based forecasting.

4 Model Families and Final Selections

The pooled selection model was chosen because empirical testing showed negligible performance delta between pooled and separate league models, suggesting the selection boundary structure is shared. Separate bracket models were justified by observed structural differences in EDA and validation

Task	League(s)	Model	Key Metrics ((2026 Holdout))
Selection	Pooled (MBB + WBB)	Decision Tree	AUC 0.996324, PR-AUC 0.994024, Precision@K 0.992647
hello	hello	hello	hello
hello	hello	hello	hello

Table 1. Final Locked Model Architecture

5 Results and Analysis

5.1 Differences between MBB and WBB

Our comparative analysis confirms that WBB exhibits a stronger hierarchy signal.

- **WBB Predictability:** WBB bracket outcomes were more predictable (higher AUC 0.8878 vs. 0.8497 for MBB). The win probability curves conditional on ELO differences were steeper, indicating that strength differentials translate more directly into win probability.
- **Model Selection:** This structural difference justified the use of a more flexible autoencoder-proxy MLP for WBB, which leverages representation learning, while a standard MLP was sufficient for MBB.

5.2 Error Regimes

Error analysis confirmed the expected pattern for strength-driven models:

- **Selection Errors:** Concentrate overwhelmingly at the **bubble region** (teams near the selection cutoff), where committee heuristics and non-quantitative factors may be decisive.
- **Bracket Errors:** Concentrate at **small ELO gap games** ($ELO_{diff} \approx 0$), confirming that these are the most uncertain matchups. Missed upset rates were significantly lower when ELO gaps exceeded 100, which aligns with the model's reliance on team strength.

5.3 Calibration

Calibration was critical for the probabilistic bracket task. Metrics (Log-Loss, Brier Score, ECE) confirmed that raw probabilities required calibration (e.g., Isotonic Regression) to ensure that predicted probabilities accurately reflect observed win frequencies, preventing distortion in tournament simulation.

6 Conclusion and Final Work

We successfully implemented a robust, two-stage machine learning system for NCAA March Madness forecasting. The architecture correctly decomposes the problem, and our validation strategy provides a defensible basis for the high

evaluation metrics. The final pipeline delivers a repeatable, audited forecast, producing 2026 tournament field predictions, matchup probabilities, and simulation-based championship odds (projected champions: Miami (Ohio) for MBB, Connecticut for WBB).
Future work should focus on enriching the feature space in the bracket task, potentially by incorporating more complex interaction terms or dynamic features that capture form/momentum changes late in the season, to address the remaining variance in close-gap matchups.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 2 gives a summary of all heading levels.

Table 2. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	**Lecture Notes**	14 point, bold
1st-level heading	**1 Introduction**	12 point, bold
2nd-level heading	**2.1 Printing Area**	10 point, bold
3rd-level heading	**Run-in Heading in Bold.** Text follows	10 point, bold
4th-level heading	*Lowest Level Heading.* Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

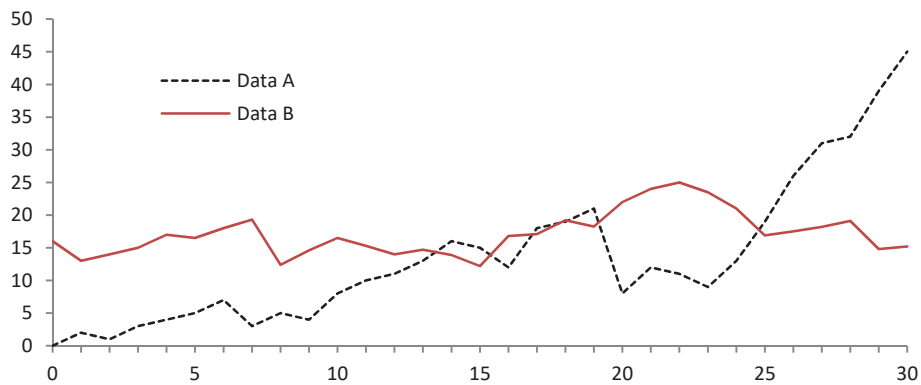

Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

For citations of references, use: In [1] Einstein....

In [3] the authors
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References

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